

Tutorial 11

Sukalyan Deb

November 2025

Non degenerate perturbation theory on two identical bosonic particle

Two identical spin-zero bosons are placed in an infinite square well . They interact weakly with one another, via the potential

$$V(x_1, x_2) = -aV_0\delta(x_1 - x_2)$$

(where V_0 is a constant with the dimensions of energy, and a is the width of the well).

1. First ignoring the interaction between the particles , find the ground state and first excited state - both wave functions and associated energies.
2. Use first order perturbation theory to estimate the particle particle interaction on the energies of the ground state and first excited state.